

Felipe Meneguitti Dias

Models, signals and software for healthcare.



I am a Senior R&D Specialist at Samsung with a PhD from the University of São Paulo. My work spans healthcare AI, signal processing and model validation, from development to deployment.

Selected projects

01 Gorgona

An ECG classification platform that connects urgent care computers to the InCor server and sends Telegram alerts to the clinical coordination team. Automated analysis runs alongside telemedicine.

Project leadership and initial system development

02 ECG classification at InCor

An ECG image classification system integrated into emergency care at InCor. A web application displays the results to help clinicians prioritize ECG review.

Model development, annotation coordination and clinical integration

03 RPMS

A remote patient monitoring system that connects wearable devices, a smartphone and a clinical dashboard. The phone processes PPG signals and sends physiological estimates to the server for patient follow-up.

Data collection protocols, signal processing and algorithm development

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ECG classification integrated into urgent care

42,037

processed records

2 min

ECG to alert in the case

41 min

before the first report

Dashboard query on October 3, 2026: all units, full period, Processed status, Current of Injury and Other classes. The count represents examination records, not unique patients.

Objective

Flag ECGs with suspected current of injury and notify the clinical coordination team earlier. Automated analysis runs alongside the telemedicine workflow. Patient assessment and care decisions remain with the medical team.

Previous workflow

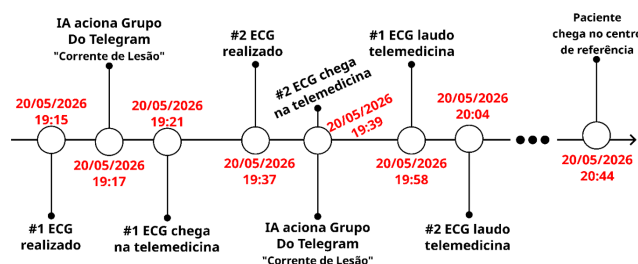
ECGs were sent to telemedicine. Reports and medical assessment informed contact with the coordination team.

With Gorgona

The Windows client detects the examination and sends it to InCor. The server classifies it and alerts the Telegram group when Current of Injury probability exceeds 50%.

I connected the client, server, web dashboard and alerts. The design includes configurable local anonymization, HTTPS and per-installation credentials.

Case study · UPA Rodrigo Argolo · 20/05/2026



Original timeline, slide 23. Personal identification block omitted.

Two ECGs from the same encounter triggered alerts within 2 minutes, preceding their reports by 41 and 25 minutes. The case documents communication timing.

The classifier was developed from the AIMED model that won the 2024 PhysioNet Challenge classification task. The CinC 2024 paper and my doctoral thesis document this foundation.

Team publications

Image-Based Electrocardiogram Classification Using Pre-trained ConvNext. Computing in Cardiology, 2024.

Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo. Universidade de São Paulo · Tese de doutorado, 2025.

Image-based deep learning for emergency electrocardiogram classification. medRxiv · Preprint, 2026.

From PACS to the emergency room screen

An ECG image classification system integrated into emergency care at InCor. A web application displays the results to help clinicians prioritize ECG review.

Objective

Support the prioritization of ECG review in the emergency room by bringing automated classification into the clinical team's workspace.

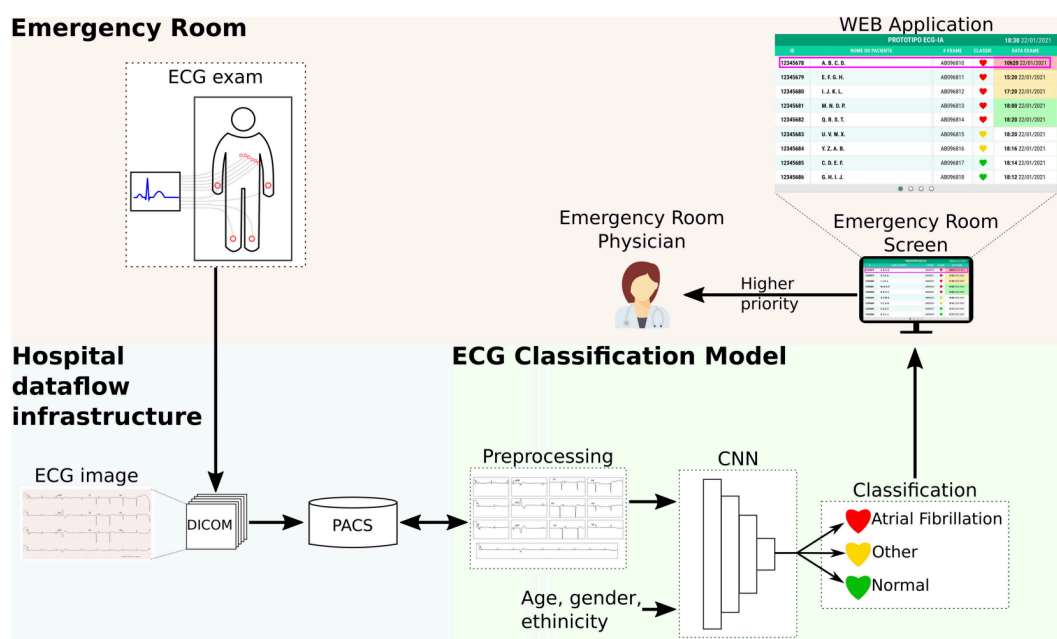


Figure 1 from Dias et al., IEEE Access (2023). Proposed integration of ECG acquisition, DICOM/PACS, classification and the emergency room web application. Image provided by the author.

In the architecture published in 2023, the 12-lead ECG is stored in DICOM format in the hospital's picture archiving and communication system (PACS). The AI module retrieves the examination, preprocesses the image and runs a convolutional neural network. The figure includes age, sex and ethnicity as classifier inputs.

The 2023 architecture uses three classes: atrial fibrillation, other and normal. Subsequent research developed a ConvNeXt ensemble for 12 categories. The 2026 study includes 18,519 ECGs from 14,402 patients and achieved a macro F1 of 0.807 on the held-out test set (95% CI: 0.788–0.825).

Model development, annotation coordination and clinical integration

Team publications

Artificial Intelligence-Driven Screening System for Rapid Image-Based Classification of 12-Lead ECG Exams: A Promising Solution for Emergency Room Prioritization. IEEE Access, 2023.

Image-based deep learning for emergency electrocardiogram classification. medRxiv · Preprint, 2026.

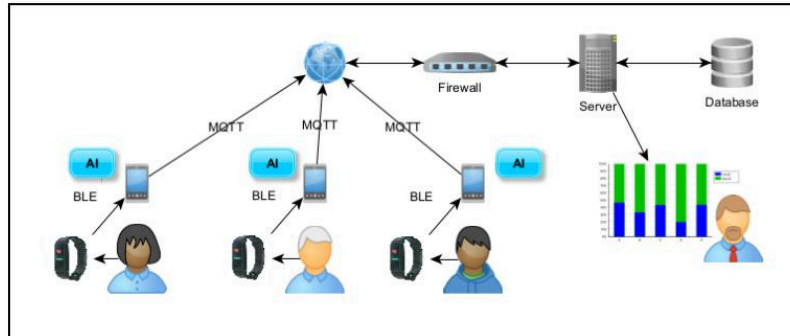
Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo. Universidade de São Paulo · Tese de doutorado, 2025.

Remote Patient Monitoring System

Support patient follow-up outside the hospital by connecting wearable recordings to biomarkers available to the clinical team.

My role

My work covered collection protocols with prototype watches, data organization, photoplethysmography (PPG) preprocessing and collaborative algorithm development.



Architecture published in Moreno, Dias et al. (SlOT, 2022), Figure 1. Signals are processed on the phone.

The watch sends PPG to the Android app over Bluetooth Low Energy. The phone processes the signal, estimates blood pressure and heart rate, and sends results over MQTT to a ThingsBoard server. The dashboard displays each patient's measurements over time.

Algorithms and research

Signal quality

Automated assessment of PPG segment quality.

Blood pressure

Calibrated estimation and evaluation of generalization limits.

Diabetes

Classification using PPG features and patient metadata.

Sleep

Sleep stage classification using temporal context.

Team publications

IoT Medical Device Architecture to Estimate Non-invasive Arterial Blood Pressure. Symposium on Internet of Things (SlOT), 2022.

Quality Assessment of Photoplethysmography Signals For Cardiovascular Biomarkers Monitoring Using Wearable Devices. arXiv · Preprint, 2023.

Exploring the limitations of blood pressure estimation using the photoplethysmography signal. Physiological Measurement, 2025.

Machine Learning-Based Diabetes Detection Using Photoplethysmography Signal Features. SBCAS, 2024.

Enhancing Photoplethysmography-Based Sleep Staging Models Through Temporal Context Optimization. Studies in Health Technology and Informatics, 2025.